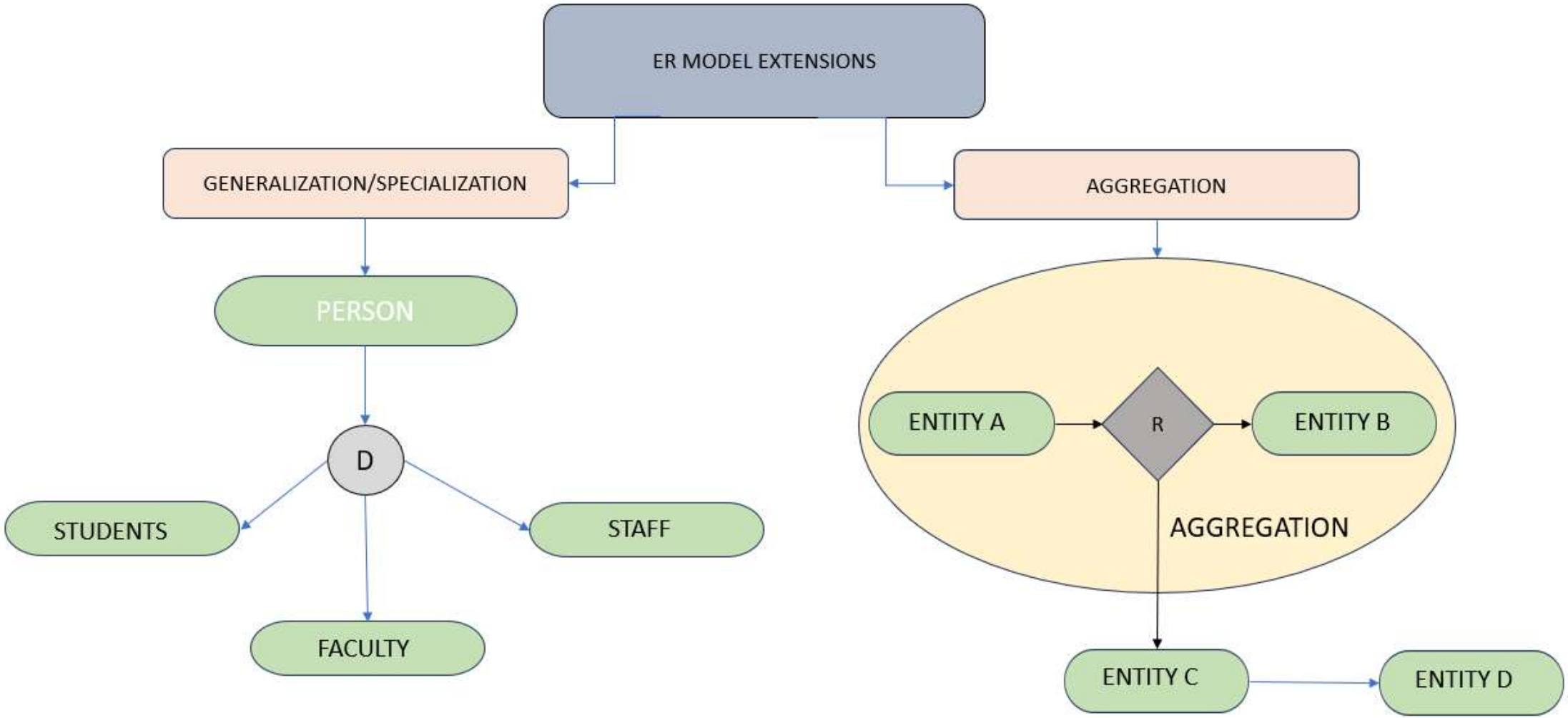
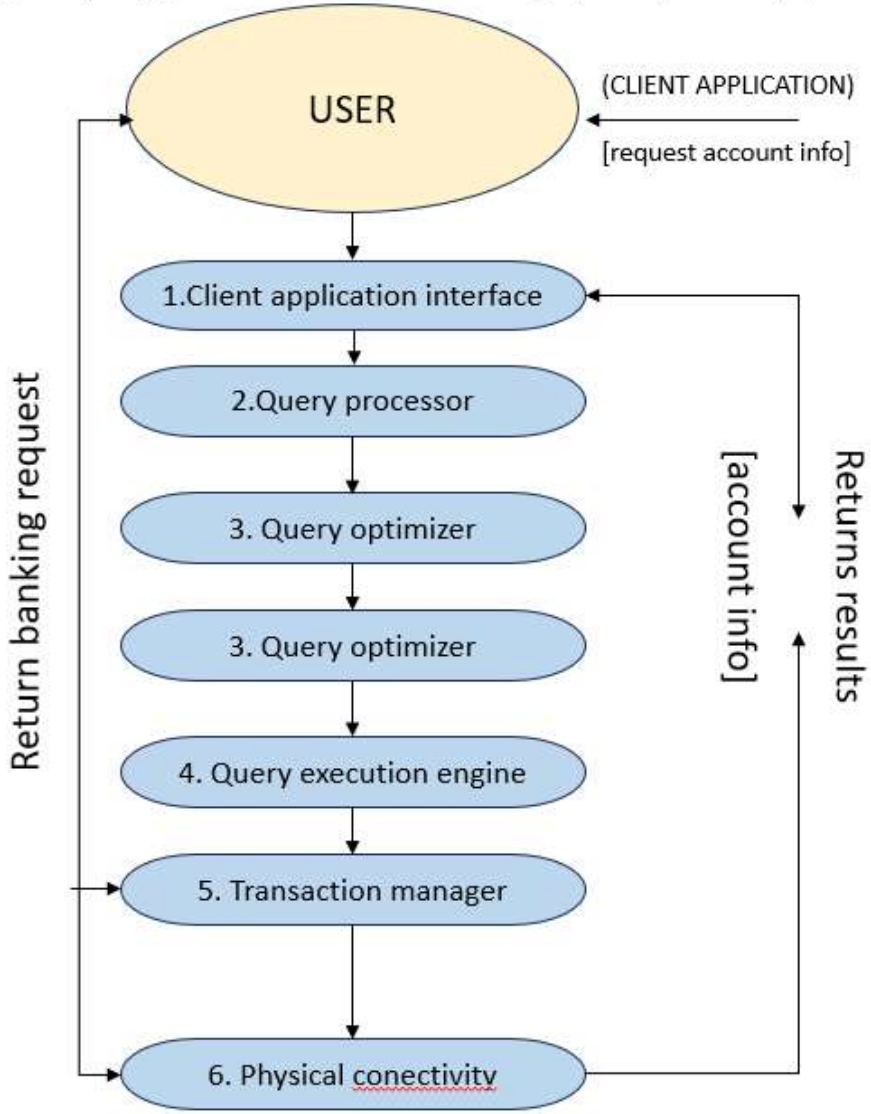
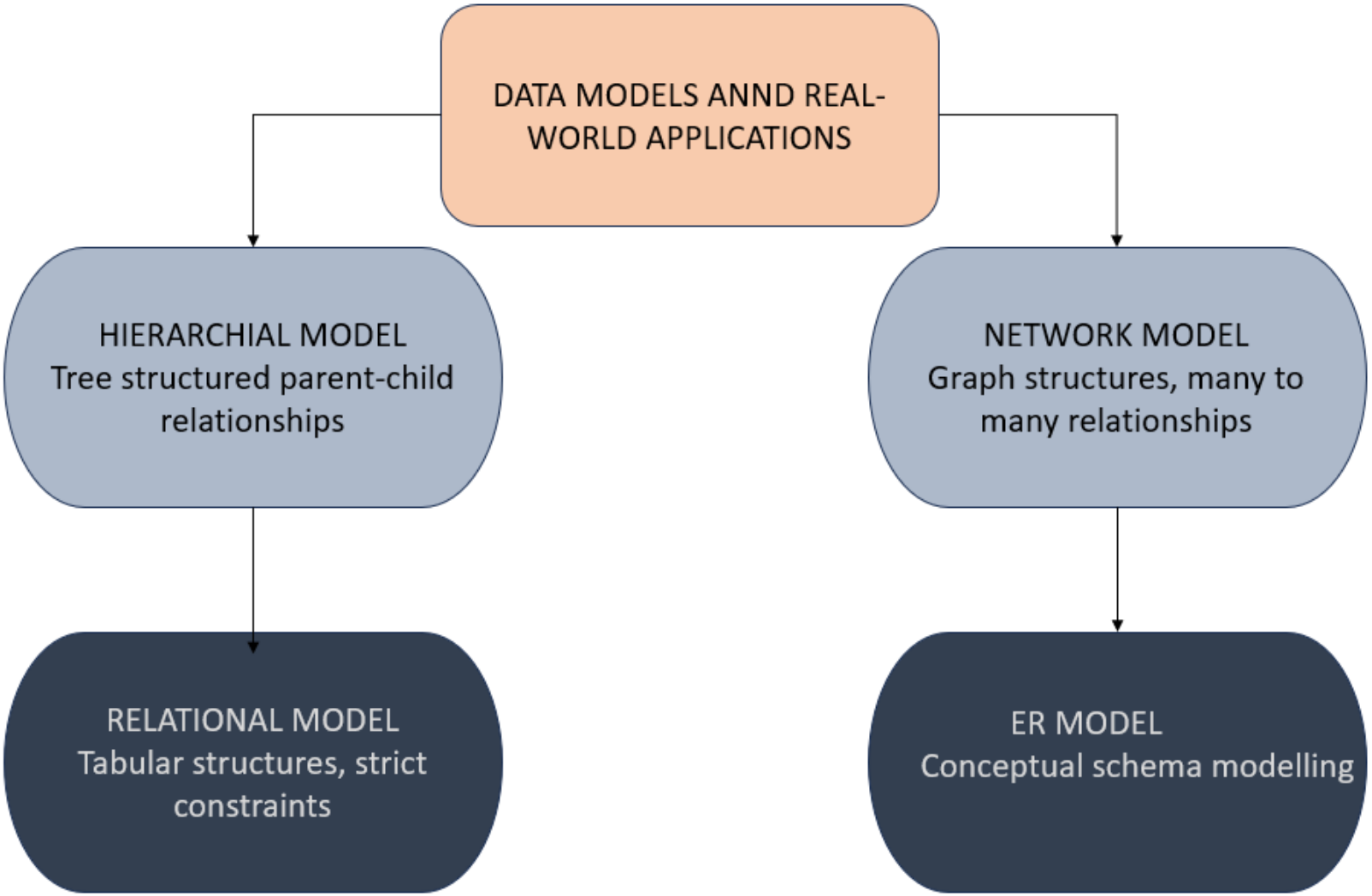
1. A university database contains Persons, Students, Faculty, and Staff, where specialized entities inherit common properties from a general entity. Explain how the EER model extends the basic ER model using generalization, specialization, and aggregation. Construct a concept map illustrating these concepts and their relationships.



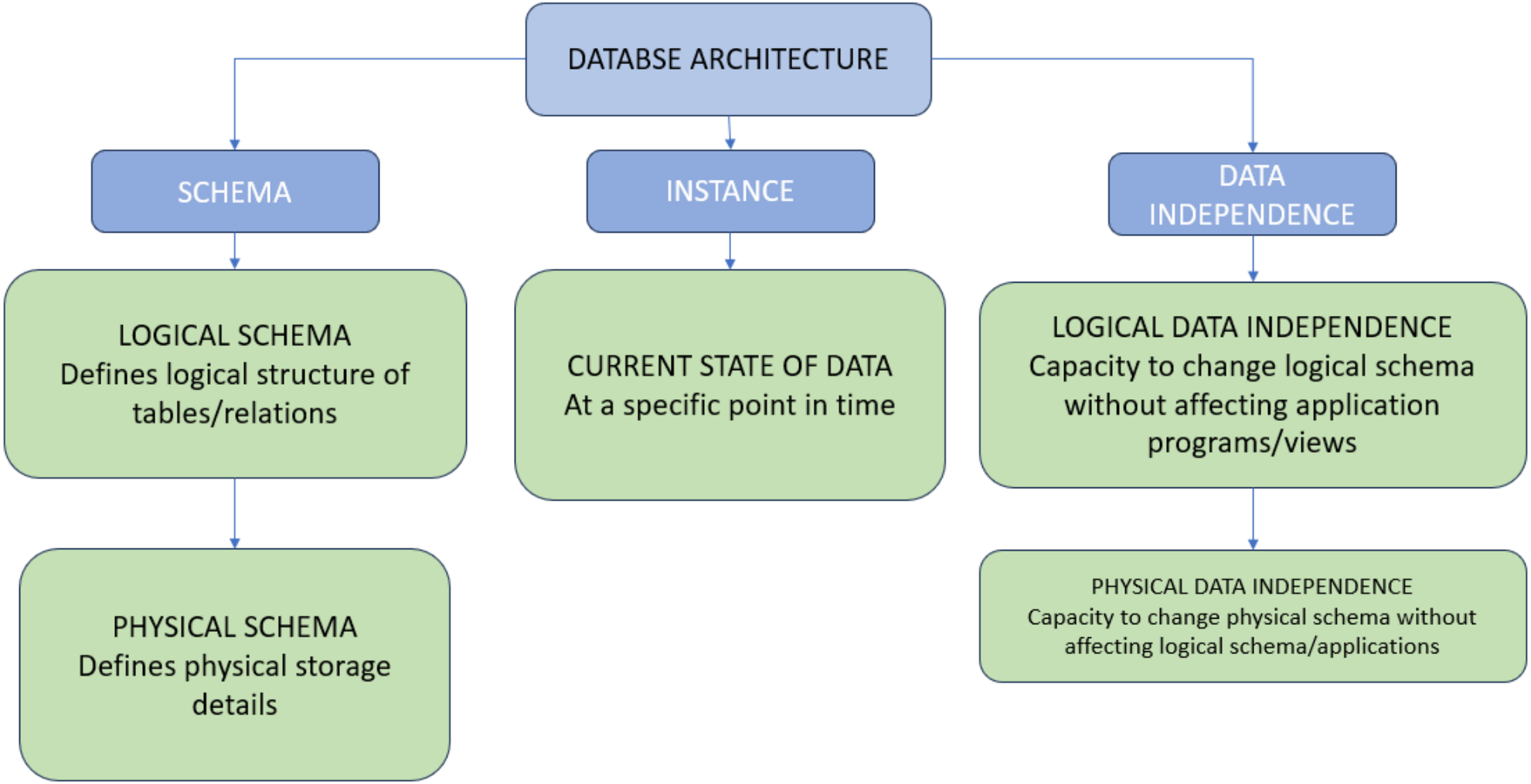
2. An online banking application receives a user's request to retrieve account information. Construct a visual concept map explaining the flow of the request through the DBMS software architecture, including the query processor, transaction manager, storage manager, database files, and data retrieval process.



3. Different organizations use different data models according to their requirements. Construct a concept map connecting Hierarchical, Network, Relational, and ER/EER models with suitable real-world applications such as organizational systems, transportation networks, banking systems, and university databases. Explain the suitability of each model.



4. A university database undergoes changes in its logical structure and physical storage without changing how users access the data. Construct a concept map explaining schema, instance, logical schema, physical schema, logical data independence, and physical data independence, and show the relationship among these concepts.



5. An organization uses a DBMS to manage its business data. Construct a concept map showing the roles of the Database Administrator (DBA), End User, and Application Developer. Include their major responsibilities and explain how they interact
within the overall DBMS environment.

